

Solvency Field Theory V12.12

The Particle:
Light, Mass, and the Machine That Makes Time
Narrative companion to the
V12.12 Particle Program

Shaun Fosmark
Independent researcher

June 2026

A particle is real energy moving at c , either open as light or closed into a self-maintaining orbit as mass, leaving permanent settlement history as it writes.

Publication boundary. V12.12 defines what a particle IS in SFT. It does not derive the microscopic settlement equation, mass eigenvalues, charge quantization, or the $6\pi \rightarrow K$ mechanism. It freezes the locally real particle picture — the one where you can close your eyes and watch the machine work — strongly enough to support the next prediction lanes.

Contents

1	One kind of stuff	2
2	Energy is gravity is compactness	2
3	The photon: open energy	2
4	The massive particle: closed energy	3
5	Formation: why energy becomes particles	3
6	No-overwrite: what it really means	4
7	Time, gravity, and expansion	4
8	What is earned and what is open	5
8.1	Earned	5
8.2	Open	5

1 One kind of stuff

There is one kind of stuff in the universe. Energy. Moving at c .

Not matter and energy as two categories. Not particles and fields as two substances. Not fermions and bosons as two tribes.

One stuff. Energy at c . Everything else — mass, charge, spin, gravity, time, expansion — is what that stuff DOES depending on how it moves.

Energy moving in a straight line is light. Energy moving in a circle is mass. Energy moving in a circle creates a clock. The clock creates time. The circling energy curves spacetime. The curvature is gravity. The accumulation of all clocks in the universe grows the boundary. The boundary growth is expansion.

One stuff. One speed. One rule: you cannot undo a completed write. Everything follows.

2 Energy is gravity is compactness

These three words do not describe three properties that happen to correlate. They describe one property measured three ways.

A photon with low energy has a long wavelength. Its energy is spread across a large spatial region. Its gravitational demand is weak. Its compactness is low. These are not three facts about the photon. They are one fact.

A photon with high energy has a short wavelength. Its energy is packed into a tiny region. Its gravitational demand is strong. Its compactness is high. Same one fact.

Turn up the energy further. The compactness increases. The gravity strengthens. At some threshold, the gravity becomes strong enough to curve the energy's own path. The photon bends. If the bending is strong enough, it closes. The energy starts orbiting. It becomes a particle.

The particle is not energy PLUS gravity. The particle is energy whose own gravity is strong enough to hold it in a circle.

And the same principle operates at every scale. Planets are matter accumulated by gravity — energy organized by compactness at planetary density. Stars are higher compactness. Neutron stars higher still. Black holes are the extreme — energy so compact that nothing can escape. The observable universe is all the energy, self-gravitating collectively at the largest possible scale.

Particles, planets, stars, black holes, and the universe share the same geometry because they ARE the same phenomenon. Energy organized by its own gravity at different scales of compactness.

3 The photon: open energy

A photon is the simplest thing energy can be. Open. Outward. Moving at c in a straight line. The entire causal budget aimed forward. No internal structure. No maintenance. No clock.

The photon does not experience time. This is not a poetic statement. It is a mathematical fact established in R181: the optical phase $\Phi = k_\mu x^\mu$ satisfies $d\Phi/d\lambda = k_\mu k^\mu = 0$ along the null ray. The photon's phase does not advance from the photon's own perspective. The frequency we measure — $\nu = E/h$ — is the rate at which wavefronts cross US, not ticks experienced by the photon.

The photon is real. It carries energy. It has momentum. It gravitates. It is not a probability cloud or a mathematical convenience. It is a physical stream of energy advancing through space.

But it has no clock because it has no closure. A clock requires a recurring event. The photon has no recurring event. It just goes. Open energy has no period, no cycle, no internal rhythm. It experiences nothing because experience requires a clock and a clock requires closure.

The photon's wavelength is its physical compactness. A gamma ray at 10^{19} eV is packed into 10^{-25} meters. A radio wave at 10^{-9} eV is spread across a meter. Same object. Same speed. Wildly different compactness. The gamma ray is close to the particle-formation threshold. The radio wave is nowhere near it.

Adding energy to a photon makes it more compact. Removing energy makes it more spread out. This is not an intensity change. It is a physical change in the spatial extent of the energy packet. The photon gets SMALLER when it gets more energetic.

4 The massive particle: closed energy

A massive particle is the same energy, no longer going straight.

At some point in the early universe — or in a high-energy collision — the energy density in a small region exceeded a threshold. The self-gravitational focusing of that energy bent its path. The bending continued until the path closed. The energy began orbiting at speed c in a circle of radius

$$r_{\text{spin}} = \frac{\hbar}{2mc}, \quad (1)$$

with the reduced Compton wavelength $\hbar/(mc)$ as the orbit diameter.

The closure creates everything that distinguishes a massive particle from a photon:

Mass. The energy is no longer propagating outward. It is circulating internally. The inertia of that circulation is what we call mass. $E = mc^2$ is not a conversion formula between two substances. It is the statement that mass IS the energy of internal circulation.

A clock. The orbit has a period — the Compton time $T_C = h/(mc^2)$. Each completed circuit is one tick. The accumulation of ticks is proper time. The photon has no clock because it has no orbit. The particle creates its own clock by closing.

Spin. The fermion branch completes a 4π spinor closure in one Compton period — two full rotations per tick. The spin angular momentum is $\hbar/2$, not inserted by hand but forced by the orbit geometry.

Gravity. The compact circulating energy curves spacetime around itself. The settlement envelope — the lag sphere — is the particle's gravitational signature.

Time dilation. Under a boost, the one-speed budget splits: $v^2 + u_{\text{internal}}^2 = c^2$. More spatial velocity means less internal circulation means slower clock. The orbit doesn't change. The allocation does.

And the orbit is rigid. The nine stable closure modes from V12.8 are nine discrete fixed points of the self-gravitational feedback. You cannot smoothly deform an electron into a slightly heavier electron. Adding energy goes into spatial motion. The internal mode refuses to absorb it because there is no nearby stable orbit to shift to. Mass is quantized because self-consistent self-gravitational orbits at speed c are discrete.

5 Formation: why energy becomes particles

Energy self-gravitates. All energy, at all scales. The question is whether the self-gravity is strong enough to close the path.

In the post-shatter universe, the energy density is enormous. Random fluctuations create regions of slightly higher density. Higher density means higher compactness. Higher compactness means stronger self-gravity. Stronger self-gravity means more focusing. More focusing means higher density. The feedback is self-reinforcing.

If the density exceeds the closure threshold for a particular mode, the energy in that region curves into a closed orbit. A particle forms. The nine stable modes are nine thresholds. The heaviest modes need the most energy density and form first, when the universe is hottest. The lightest modes form last.

A single isolated photon cannot spontaneously become a particle. Conservation forbids it: a single null four-momentum has $p^2 = 0$, while a massive particle has $P^2 = m^2c^2 > 0$. Formation requires a physical encounter — two non-collinear photons meeting with enough combined energy, or an interaction vertex where energy can be rerouted from open to closed.

The transition from open to closed is discontinuous. You cannot smoothly dial a photon's wavelength toward the Compton wavelength and watch it gradually become a particle. The open branch and the closed branch are topologically different. Formation is a discrete event: the path closes or it doesn't.

Everything that didn't close remains as photons. The cosmic microwave background is the leftover energy from the early universe that never found a stable orbit. It has been propagating at c for 13.8 billion years, still open, still experiencing no time, still in the most primitive form energy can take.

6 No-overwrite: what it really means

The founding rule of SFT is that a completed write cannot be undone. But V12.12 corrects what “completed write” means.

The old intuition was spatial: the helix cannot cross its own past path. R182 proved this cannot be the rule. A continuous path has neighboring points that are arbitrarily close to its past, and the closed massive orbit intentionally revisits the same spatial coordinates every Compton period. If no-overwrite meant spatial non-intersection, the massive particle could not exist.

The corrected rule is: a completed physical settlement write leaves permanent history, and the same full event cannot be written again. The spatial coordinates can recur. The event address cannot.

The particle orbits. It returns to position $\mathbf{x}$ after one Compton period. But $\mathbf{x}$ at time $t + T_C$ is not the same event as $\mathbf{x}$ at time t , because the history count N has advanced. The old write is preserved. The new write requires new capacity. Expansion provides that capacity.

No-overwrite and expansion are therefore the same mechanism viewed at different scales. The particle cannot write into already-settled history. The universe grows to provide room for the next write. The permanence of the past and the growth of the future are two descriptions of one constraint.

7 Time, gravity, and expansion

One process. Three scales. Three names.

At the particle scale: the closed orbit repeats. Each circuit is a tick. The accumulation of ticks is proper time. Time is the maintenance cadence of closed energy. The photon does not experience time because it is not closed. The massive particle creates time by existing.

At the regional scale: compact energy curves spacetime. The settlement geometry responds to the density of writes. More writes per volume means stronger curvature. Gravity is not a force pulling objects together. It is the geometric consequence of energy compactness. Objects “fall” because the time direction is curved toward higher energy density, and everything is already moving through time at c .

At the cosmic scale: every particle tick adds capacity to the boundary. The total write rate of all particles drives the growth of the boundary area. The boundary growth is the expansion of space. $H = \dot{S}/I$ — the expansion rate equals the total write rate divided by the total information already written.

The arrow of time, the attractive nature of gravity, the expansion of the universe, and the second law of thermodynamics are four consequences of one rule: writes are permanent. You cannot erase a tick. You cannot uncurve settled geometry. You cannot shrink the boundary. You cannot decrease the total information. One constraint. Four phenomena.

8 What is earned and what is open

8.1 Earned

The particle is real energy at c . Light is open. Mass is closed. The photon has no clock. The massive particle creates its clock by closing. The fermion orbit has radius $\hbar/(2mc)$ with 4π spinor closure. The one-speed budget $v^2 + u^2 = c^2$ closes exactly. Mass is quantized because self-consistent orbits are discrete. No-overwrite is settlement permanence, not spatial avoidance. Energy is gravity is compactness at every scale.

8.2 Open

The microscopic settlement equation. The native self-focusing action. Charge quantization. The $6\pi \rightarrow K$ mechanism. Photon support and localization. Interaction and decay rules. The radiation contribution to expansion. The full species map from modes to particles.

Each of these is a specific target. The foundation is strong enough to support their pursuit. The particle is defined. The predictions can begin.

Acknowledgments

The R156–R185F particle audit was developed and executed in collaboration with ChatGPT/5.5 (OpenAI). The locally real particle picture, the energy-is-gravity-is-compactness principle, and the reality-lock correction were developed with Claude/Opus (Anthropic). The 6π invariance audit that triggered the particle program was identified by Claude Mythos (Anthropic).

References

- [1] S. Fosmark, *Solvency Field Theory V11: Master Framework*, Zenodo (2026).
- [2] S. Fosmark, *SFT V12.9: Mass as Geometric Focusing*, Zenodo (2026).
- [3] S. Fosmark, *SFT V12.11: Master Action Audit and Upgrade*, Zenodo (2026).